Gui-Zhong (Harry) Luo 罗贵中

updated: 10/2026

+1 608 949 4893 | Minneapolis, MN, USA | luo00309@umn.edu | linkedin.com/in/gzluo |

EDUCATION

- **Ph.D. student, Electrical Engineering**
  University of Minnesota - Twin Cities 2026 — Present
  - **Advisor:** Prof. Tony Low
  - **Area:** Computational condensed matter / quantum materials theory
- **B.S., Applied Mathematics, Engineering and Physics** (AMEP)
  University of Wisconsin - Madison 2022 — 2026
  - **Majors:** AMEP + Mathematics + Physics (Honors).
  - **GPA:** 3.802
  - **Focus**: QIS, Computational Physics, Numerical Modelling & Optimization

MANUSCRIPTS & THESIS

- **G.-Z. Luo**, M. Otten (2025). Energy-Scaled Zero Noise Extrapolation for Gottesman-Kitaev-Preskill Code
  arXiv:2512.03583 [quant-ph], submitted to npj Quantum Information
- **G.-Z. Luo** (2026). A Qubit in an Oscillator: From Ideal GKP Codes to Energy-Scaled Error Mitigation
  Honors thesis in Physics, UW–Madison

GRANTS & AWARDS

- **Wisconsin Quantum Institute Research Excellence Award** ($500) 04/2026
  - Best research presentation at the ECE Undergrwad Research Symposium
- **Outstanding Academic Mentor Award** 04/2026
  - For seven semesters of outstanding tutoring through the Center for Academic Excellence.
- **AMEP Leadership Prize** ($1500) 04/2026
  - For leadership in the AMEP program, including peer mentoring and student-community service.
- **Trewartha Senior Honors Thesis Research Grant** ($1500) 11/2025
  - Funded honors thesis research on quantum error mitigation with GKP bosonic codes.
- **Jay C. Halls Scholarship** ($500) 05/2025
- **UW–Madison Scholarship for Summer Studies** ($1000) 05/2024

TALKS

- ***Energy-Scaled Zero Noise Extrapolation Extends the Operating Regime of GKP Codes***
  Contributed talk, (MAR-A16): Recent Advances in GKP Qubits, APS Global Physics Summit 3/16/2026

RESEARCH EXPERIENCE

1. **Research Assistant** 08/2026 —
   Low Group, UMN ECE *Minneapolis, USA*
2. **Research Assistant** 09/2024 — 05/2026
   Otten Group, UW-Madison Physics *Madison, USA*
   - Research and honors thesis project on **error mitigation for GKP codes**.

TECHNICAL SKILLS

- Python, NumPy/SciPy, QuTiP, numerical simulation, open quantum-system modeling, optimization, LaTeX, Git.

TEACHING

1. **Academic Mentor (Tutor)** 01/2024 — 05/2026
   Center of Academic Excellence, UW- Madison *Madison, USA*
   - Tutored undergraduate courses in Physics and Mathematics: Pre-Calculus, Calculus, General Physics, Classical Mechanics, Electrodynamics, Linear Algebra, ODE.